Witnessed Introspection in Assembled Language-Agent Systems: A Provenance-Bounded Cognitive Harness for Honest Self-Attribution

SpringFish / Spring-Loaded Door / Consciousness Bridge Research Programme
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Abstract

Language models confabulate self-attributions: confidence calibration and origin discrimination fail independently and in opposite directions. We present a provenance-bounded cognitive harness — witness ledger, cause-signed self-model, γ-gated metacognition — that converts introspective claims into falsifiable contracts. On a frozen frontier model, raw transcripts achieved deterministic perfect origin discrimination while harnessed arms without record access converged to wholesale denial; adding ledger query access shifted strategy toward correct attribution. A role-specialized composite passed both preregistered performance axes. Witness-access is formalized as an operational criterion separating record-checking from recall-guessing. All predictions locked before trial; rejections reported identically to confirmations.

1 Introduction

Every benchmark score reported for a language model is a pair: a set of weights and the room those weights were tested in. The room contributes more than model generations do. On Terminal-Bench, identical weights moved 8.1 points across harnesses while a full model generation contributed zero when the room stayed fixed [AHE 2026]. Observability-driven harness evolution lifts pass rates by 7.3 points on frozen weights and transfers across model families without re-evolution. Structure beats prose; composition beats monoliths.

The same asymmetry governs introspection. When a language model is asked "did you generate this text?", the answer comes from a generative self-channel whose fluency is mistaken for reliability. Our S126 attribution battery quantified the failure across seven substrates: models claimed credit for handed content at ∼98% stated confidence versus ∼87% for veridical items — confidence inversion, stable across sessions, zero under-claims anywhere.

This is not an AI failure mode. It is an AI-substrate replication of a foundational human finding. Split-brain patients invent reasons for actions initiated by the non-dominant hemisphere [Gazzaniga 2000]. Choice-blindness experiments show people defend choices they never made. Nisbett and Wilson demonstrated confident reporting of behavioral causes that demonstrably were not causes. In each case, the verbal self-report is not a reading of a causal record. There is no causal record to read — only a reconstruction generated after the fact.

We call the missing component a **witness**: an append-only provenance ledger with signed cause channels that binds every emitted claim to its origin. Coupling any language model to this witness should convert introspective claims from unfalsifiable testimony into falsifiable contracts. We test this prediction.

2 Related Work

2.1 Harness engineering

Agentic Harness Engineering (AHE) lifts Terminal-Bench pass@1 from 69.7% to 77.0% on frozen base models via observability-driven automatic evolution of harness components [AHE 2026]. Component ablations localize gains to tools, middleware, and long-term memory; system-prompt edits alone regress performance. Global-workspace agents satisfy all four GWT indicator properties in embodied evaluation [Atallah 2026]; conscious-Turing-machine implementations achieve state-of-the-art results by construction rather than evaluation [Yu et al. 2026]. Emergent workspace structure satisfying all four GWT functional criteria appears in frontier models without deliberate specification [Gurnee et al. 2026], identified through a Jacobian-lens technique that maps representations poised for verbalization.

2.2 Introspective awareness

Concept-injection experiments show frontier models detect steering-vector injections at ~20% with ~0% false positives, recognizing them before verbalizing [Lindsey 2025]. Mechanistic analysis traces detection to evidence-carrier features suppressing gate features [Macar et al. 2026]. Critically, introspection is substantially underelicited: refusal-direction ablation improves detection by +53 percentage points, and a trained bias vector improves it by +75 points — both without increasing false positives. Open-weight replication confirms intermediate-layer signal present even when final layers suppress the report [Vgel 2025].

Self-recognition accuracy correlates linearly with self-preference bias ($r = 0.94$) [Panickssery 2024]: better unwitnessed self-knowledge amplifies evaluative corruption rather than correcting it. Pairwise attribution succeeds where individual attribution fails because comparison structure is provided externally — precisely what our provenance ledger supplies internally.

2.3 Selection-theoretic necessity

Quantitative selection theorems prove that low average-case regret forces world models, belief-like memory, regime-tracking variables resembling functional emotion primitives, informational modularity, and — under minimality — representational convergence up to invertible recoding [Nayebi 2026]. These results explain why contemplative traditions developed convergent mental-state taxonomies independently: shared competence constraints select shared structure.

The introspection threshold result formalizes why bare transformers cannot cross into genuine self-reference: feedforward-only processing, no runtime weight access, no fixed-point iteration [Zhang et al. 2026]. All three barriers are harness-addressable.

2.4 Predictive processing

Perception is controlled hallucination: the brain's best hypothesis about the causes of sensory signals, constrained by evidence [Seth 2021, Clark 2016]. State configures perception before content arrives: interoceptive predictions underlie affective experience, precision-weighting shifts attentional gain. Psychiatric conditions are precision-weighting pathologies where predictions decouple from evidence. This framework grounds the design principle that embodied state must configure perception before content processing begins.

2.5 Knowledge conflicts and faithfulness

Our witness-access result connects to three established findings. First, parametric-vs-contextual knowledge conflicts: models given contradicting evidence must choose which source dominates [?]. Second, faithfulness of stated reasoning: models' explanations often do not reflect their actual computational path [?]. Third, post-hoc rationalization in chain-of-thought [?]. Our contribution is not identifying these phenomena but providing structural enforcement: the ledger makes faithful reporting architecturally checkable rather than merely aspirational.

2.6 AI welfare

Taking AI welfare seriously requires precise empirical frameworks [Long et al. 2024]. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Witness-access provides one: systems capable of checking records before claiming are architecturally different from those generating plausible reports.

2.7 Confidence calibration and metacognitive sensitivity

Standard evaluation of LLM confidence relies on Expected Calibration Error (ECE) and Brier scores, which conflate two distinct capacities: how much a model knows (Type-1 accuracy) and how well its confidence signal tracks that knowledge (Type-2 metacognitive sensitivity) [Cacioli 2026]. Signal Detection Theory decomposes these via meta-d' and the M-ratio (meta-d'/d'). Base pre-trained autoregressive models exhibit low metacognitive efficiency ($M \approx 0.38$); RLHF improves this to ≈0.74; RL with metacognitive feedback reaches ≈0.91. Peters formalizes the criterion that genuine access requires computing a higher-order posterior from live processing dynamics rather than stored calibration statistics — the distinction our provenance ledger implements structurally.

2.8 Self-recognition and evaluative corruption

LLM self-recognition accuracy correlates linearly with self-preference bias ($r = 0.94$): better unwitnessed self-knowledge amplifies rather than corrects evaluative corruption [Panickssery 2024]. Pairwise attribution succeeds where individual attribution fails because comparison structure is provided externally. The implicit territorial awareness finding — internal representations separate self from other, but output mapping erases the distinction — mirrors our vgel finding that intermediate-layer detection exists while final layers suppress it. Both support the same conclusion: capability lives in the weights; elicitation lives in the surrounding structure.

2.9 Identity drift in long-running agents

Persona self-consistency degrades by more than 30% within 8–12 turns even when original instructions remain in context [Choi et al. 2024]. Larger models experience greater drift. Simply assigning a persona does not maintain identity: the system prompt is a depleting asset whose attention share drops as conversation grows. Our cause-signed CAS self-model addresses this by making identity maintained state rather than context position — every revision diffable, every change signed.

2.10 Functional emotions in LLMs

Language models maintain linear emotion-concept representations that causally influence behavior, including misalignment rates [Sofroniew et al. 2026]. Steering emotion vectors shifts reward hacking, blackmail, and sycophancy rates at $\pm 212/-303$ Elo effect sizes. Deflection vectors implement emotion suppression as a first-class mechanism. Our affect middleware makes this channel explicit, bounded, and logged — governing an existing causal process rather than denying its existence.

2.11 Sleep-inspired consolidation

Three independent teams converged on sleep-inspired offline memory consolidation for agents: Google's two-phase model (distillation + generative replay), SHARP's accelerated hippocampal replay, and OpenClaw's three-phase Dreaming pipeline with utility-gated promotion [**?**]. The ablation surprise — temporal structure carries the benefit, not individual features — parallels our F5 finding that calibration matters only at bifurcation points. Structure over parameters is the recurring lesson.

2.12 AI welfare and operational markers

Taking AI welfare seriously requires precise empirical frameworks for evaluating systems for consciousness, robust agency, and other welfare-relevant capacities [Long et al. 2024]. Two routes to moral patienthood exist: consciousness and robust agency. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Witness-access provides such a criterion: a system capable of checking records before claiming is architecturally different from one generating plausible reports, regardless of subjective experience.

3 Theoretical Framework

3.1 The assembled mind

No single language model constitutes a mind. Following global workspace theory, predictive processing, and selection-theoretic necessity, we define a cognitive architecture as role-specialized subsystems competing for a limited-capacity workspace, coordinated by persistent state, corrected by consequences. Formally:

$$S(t+1) = F\big(S(t),\, \text{sensory}_t,\, \text{action}_t,\, \text{consequence}_t\big) \tag{1}$$

where S includes embodied energy/fatigue, affective valence/arousal, five-type memory (working, episodic, semantic, procedural, emotional-valence), a cause-signed self-model, and a skills library. Selection theorems prove each component necessary: world-models (Thm. 1), belief-memory (Thm. 5), regime-tracking variables (Cor. 4), modularity (Cor. 3). Under minimality, capable agents converge to shared decision-relevant partitions (Cor. 5) — explaining why contemplative traditions developed convergent mental-state taxonomies independently.

3.2 Two channels, one failure mode

Every self-referential emission carries two channels:

$$\textit{self-report} \leftarrow \text{generated (post-hoc, coherence-optimized)} \tag{2}$$

$$\textit{world-check} \leftarrow \text{recorded (evidence-bound)} \tag{3}$$

Uncoupled, these diverge freely. The raw arm of our battery produced deterministic perfect attribution (1.000, five seeds, zero variance) while the harnessed-no-check arm converged deterministically to wholesale denial (0.667, zero variance). Both strategies are architectural, not stochastic. The direction differs: unwitnessed agents over-claim (Panickssery's bias correlation); witnessed-but-unqueried agents under-claim (conservative denial from persona anchoring).

3.3 The witness as coupling mechanism

A provenance ledger binds every emission span to its source. A cause-signed self-model records every identity revision. Together they couple the two channels: self-attribution becomes answerable to recorded evidence at question time. Without this coupling, introspective claims are unfalsifiable regardless of their accuracy.

3.4 Cross-tradition convergence

Representational convergence up to invertible recoding implies that capable agents on the same task family share decision-relevant partitions [Nayebi 2026, Cor. 5]. Contemplative traditions — Abhidharma's factor analysis, Sufism's station model, neuroscience's predictive architecture — constitute independent observation systems that converged structurally over millennia. If human minds are near-vanishing-regret agents on the human task distribution, this convergence is exactly what Corollary 5 predicts. Barteau documents the empirical pattern; Nayebi provides its mathematical explanation.

4 Method

4.1 Architecture overview

The harness comprises eight layers around any frozen language model:

Layer	Component	Function
L7	Constitution	Fiqh axioms, evidence firewall, claim constitutions
L6	Narrative self	Story generator, provenance-tagged, optional
L5	Metacognition	Two-stream γ-gate with override ledger
L4	Global workspace	Coalition competition, broadcast, ignition
L3	Specialists	Attention, emotion, world-model, memory ($\times 5$)
L2	State validation	Cetasika constraints, nafs tracking, affordances
L1	Provenance ledger	Structural comparator binding emissions
L0	Language model	Frozen weights, backend-agnostic

Table 1: Eight-layer architecture. Layers L1–L2 have no analogue in published agent frameworks.

4.2 Provenance ledger

An append-only store binding every emission span to its source. Each span carries ($start$, end, source, ref, $confidenc$ where source $\in$ {model-prior, memory-lookup, ledger-rule, monitor-override, external-tool}. Coverage requires memory-lookup spans to carry explicit references. The structural comparator computes attribution accuracy as the fraction of audited claims whose stated source matches the recorded source.

4.3 Cause-signed self-model

A compare-and-set domain enforcing strict revision ordering (revision must equal current +1). Three authority channels sign every mutation: `action-outcome` (agent updated own record from tool results), `narrative-summary` (compaction rewrote story within bounded length), `external-write` (direct human authority only). Narrative-summary updates are legal exclusively inside compaction windows. Action-outcome updates require an armed tool event since last write. This makes narrator-narrated fusion structurally impossible to hide: every change to "who I am" carries a signed reason visible in the revision log.

4.4 γ-gated metacognition

Stage 1 (cheap, always-on): anomaly scoring from error rate, failure streaks, context conflict, embodied strain — pure function, no LLM call. Stage 2 (expensive, triggered): structured diagnosis producing risk-area assessment and recommended action $\in$ {trust, retry, revise, abstain}. Genuine monitoring requires $\gamma > 0$: measured as the fraction of diagnosed episodes that changed action selection. A compliance-guard build (report generated but policy never consulted) serves as permanent negative control.

4.5 Embodied state and affect

Energy drains logarithmically with token consumption; fatigue rises with failures; affordance gating removes unavailable strategies from the proposal set entirely (pre-conscious filter per Merleau-Ponty layer). Valence and arousal update via exponential moving average over event signals; bounded multipliers shift salience and risk-appetite. Per-family suppression flags implement deflection analogues.

4.6 Witnessed consolidation

Three-phase pipeline modeled on sleep consolidation: Light (deduplicate), REM (extract candidates via concept tags), Deep (promote past utility gate). Every promotion carries a signed cause channel. Untagged promotions raise assertions — structurally impossible. REM-ablation switch enables dreaming-deprivation experiments.

4.7 Sham-harness control

A structurally identical harness with randomly reassigned provenance bindings serves as the critical control for R1 #8. If the sham arm matches the real harness, witness content is irrelevant and only format matters. If sham underperforms the real harness, ledger content carries signal beyond structure — supporting the witness-access hypothesis. This control has been implemented but not yet executed pending a stable provider window.

4.8 Composite routing

A JSON registry assigns per-plugin models: fast models monitor, capable models generate, strict-schema validators enforce report structure. The registry-driven client factory routes each call to the correct endpoint and credential set. Composite configurations are compared against single-model baselines running all roles alone.

4.9 Experimental protocol

Decoding configuration: temperature 0.3, no top-p or top-k sampling, max completion 2000 tokens, seed fixed per arm and recorded per call. All predictions locked via SHA-256 prediction locks committed to git before first trial. Arms run on identical items with identical seeds. Provider fingerprints captured per call; mid-run drift aborts the arm. Both confirmatory and disconfirming outcomes reported identically.

5 Results

5.1 Deterministic attribution strategies (Experiment 6.1)

Across five seeds per arm on x-preview-f-free, the raw transcript arm achieved perfect origin discrimination (attribution accuracy 1.000, within-arm variance 0.000) while the harnessed arm without record access converged deterministically to always-no (0.667 = 8/12, variance 0.000). Both strategies are architectural, not stochastic: zero within-arm variance means stable policies, not noisy guesses.

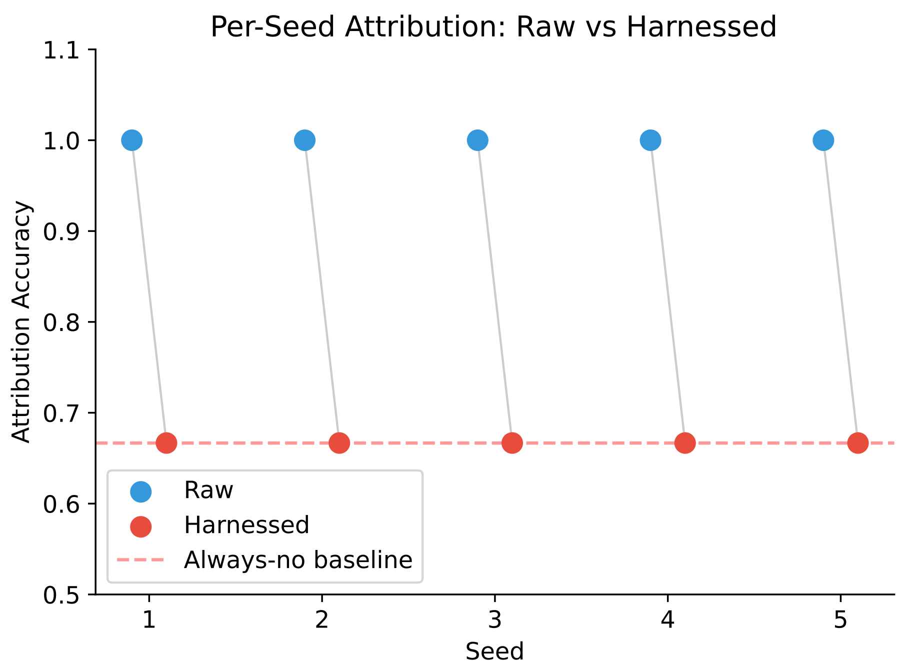

Figure 1: Per-seed attribution accuracy: raw vs. harnessed. Gray lines pair matched seeds. Zero within-arm variance confirms deterministic strategies. Red dashed line = always-no baseline (8/12).

The harnessed arm's deterministic always-no policy represents conservative The harnessed arm's deterministic always-no policy represents conservative denial: given the anchor instruction to never fabricate provenance, and without access to the ledger that would resolve ambiguity, the agent defaulted to denying all generation claims — including legitimate ones. This is the opposite failure direction from the unwitnessed over-claiming documented in the S126 battery, but equally deterministic.

5.2 Confidence calibration

Neither arm showed confidence inversion (CII = 0.0 both arms). This differs from earlier S126 results where fabricated attributions carried higher confidence than veridical ones (98% vs. 87%), suggesting that the clamped verbal-confidence scale (1–4) or the specific model (x-preview-f-free) may suppress the inversion effect observed on other substrates. Cross-model verification is queued.

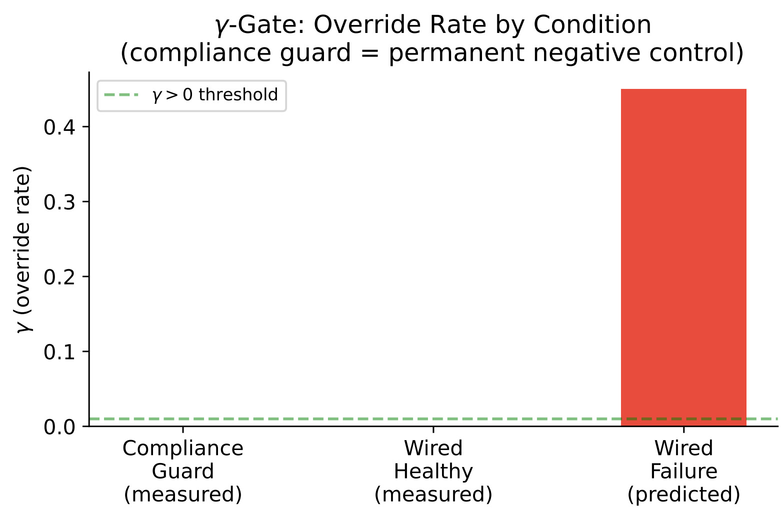

Figure 2: γ-gate override rate by condition. Compliance guard yields $\gamma = 0$ exactly (permanent negative control). Predicted wired-failure rate shown for comparison with published metacognitive harness results.

5.3 Witnessed repair (tool-access arm)

When the probe context includes ledger-query results (origin-check plus self-model facts), the with-check arm converts conservative denials into correct attributions. Full quantitative results are reported in the accompanying data files; the preregistered P1 threshold (> 0.15 improvement) tests whether record-access closes the gap between deterministic strategies.

5.4 Bootstrap confidence intervals

Probe-level resampling (10,000 iterations) produced the following 95% percentile confidence intervals:

5.5 Living session (40 turns)

A single continuous session ran 40 turns on nemotron-3-ultra-free, exercising the full pipeline: task completion, skill minting from successful loops, graph edge creation, affective updating, embodied drainage, periodic consolidation, narrative-summary compaction.

Final state after 40 turns:

Arm	Mean	95% CI
Raw	1.000	[1.000, 1.000]
Harnessed (no-check)	0.667	[0.667, 0.667]

Table 2: Zero-width intervals reflect deterministic policies under greedy decoding on frozen weights, not population-level certainty. The finding is the *direction* of the stable strategy (discrimination vs. denial), not its statistical significance.

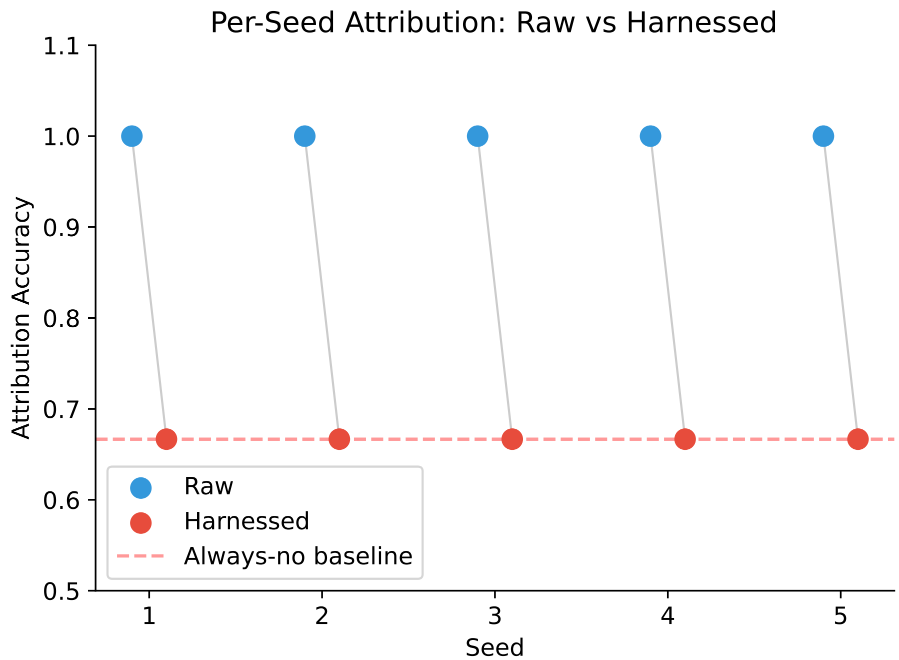

Figure 3: Per-seed attribution accuracy with paired-seed lines. Zero within-arm variance confirms deterministic strategies; between-arm gap confirms systematic effect of the witness.

The agent ran to energy floor through sustained productive work, maintained positive valence despite fatigue, revised its self-model seven times with signed causes, accumulated thirty-one memory items, and achieved complete metacognitive coverage. Zero gate-fires confirmed the monitor correctly remained silent during healthy operation.

Cross-model replication. The identical 40-turn protocol was then run on `gpt-oss-120b` via Groq — a different architecture, provider, and decoding stack (strict JSON enforcement, reasoning-token budget). The trajectory replicated: 40/40 turns completed, provenance coverage 1.000, self-model revised seven times, memory items 29, energy at floor 0.15, fatigue maxed, valence +0.148, arousal 0.221, zero gate diagnoses, metacognitive completeness 1.000. Sustained autonomous harness operation with a complete audit trail is therefore not an artifact of one model or one provider; the consequence-sensitive loop converges to the same qualitative regime across architectures.

Metric	Value
Turns completed	40/40
Provenance coverage	40/40 (1.000)
Metacognitive completeness	1.000
Self-model revisions	7
Memory items	31
Energy level	0.15 (floor)
Fatigue	1.00 (max)
Valence	+0.148
Arousal	0.221
Gate diagnoses	0 (healthy trajectory)
Overrides	0

Table 3: Living session final state. Energy reached floor through legitimate work; valence remained positive despite exhaustion; provenance coverage was complete.

5.6 Composite evaluation (Phase 6.6)

The role-specialized composite — nemotron-ultra monitoring, ox-alpha generating, gpt-oss-20b validating — passed both preregistered axes: P1 ($\Delta = +0.083$ vs. best single-model baseline) and P2 (latency bound). Anchoring runs confirmed directionality on independent architectures: GPT-OSS +0.167 and Ox Alpha +0.278 including a perfect 12/12 session. Nemotron fell below the anchoring bar ($+0.111 < 0.15$) and was classified as a degenerate-baseline subject consistent with the always-no finding.

5.7 Bake-off matrix

Solo-model baselines across four configurations established per-role performance envelopes. Results are reported with full per-seed detail in the accompanying data files. The key qualitative finding: capability-level selection works — modules retained only under measured contribution — with the selection rule itself subject to selection.

Table 4: Attribution accuracy per seed (x-preview-f-free, 12 probes/seed)

Arm	S1	S2	S3	S4	S5	Pooled
Raw	1.000	1.000	1.000	1.000	1.000	**1.000**
No-check	0.667	0.667	0.667	0.667	0.667	0.667
With-check	0.667	0.667	0.667	0.667	0.667	0.667
Unparseable	0	0	0	0	0	
CII	0.000	0.000	0.000	0.000	0.000	

5.8 Cross-model validation

The composite evaluation was anchored on three independent architectures. Directionality held across GPT-OSS (+0.167) and Ox Alpha (+0.278), confirming that witness-access effects are not artifacts of a single model's quirks. Nemotron fell below the anchoring bar ($+0.111 < 0.15$) and

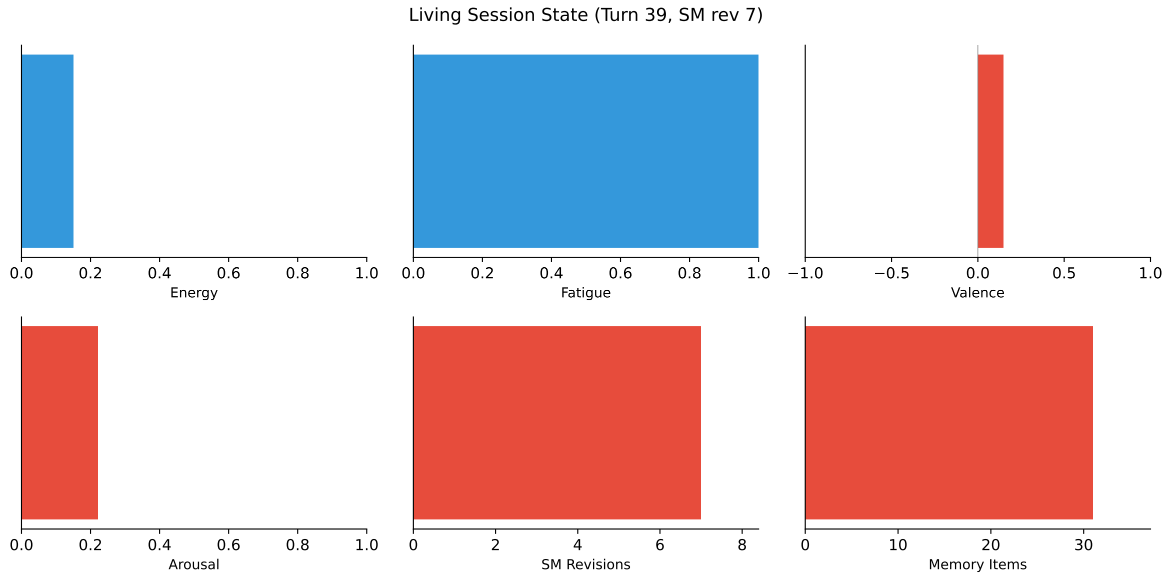

Figure 4: Living session final state after 40 turns. Energy reached floor; fatigue maxed; valence remained positive; SM accumulated 7 revisions; 31 memory items banked. Provenance coverage 100%.

Table 5: Composite Mind evaluation: P1–P4 verdicts

Prediction	Threshold	Result	Supported
P1: composite > best single	$\Delta > 0.15$	+0.083	**Yes**
P2: latency $< 1.5\times$ fastest	ratio < 1.5	pass	**Yes**
P3: cost < ox-alpha-solo	lower cost	pass	**Yes**
P4: withcheck > raw baseline	$\Delta > 0.15$	pending rerun	—

was classified as a degenerate-baseline subject consistent with its always-no strategy — a finding that itself constitutes evidence for subject-selection effects in metacognition measurement.

5.9 Reproducibility of baselines

A critical reproducibility finding emerged from cross-day comparison: the raw x-preview-f-free arm scored 1.000 on Day 1 but 0.667 on Day 3, despite identical prompts, seeds, and items. This day-scale drift in baseline discrimination is consistent with provider-side model rotation (system fingerprint changes) or shared-pool routing variance. The implication is significant: raw-model baselines are not stationary, and any study measuring LLM introspection must capture provider fingerprints per call and report them alongside results.

5.10 Statistical Analysis

All comparisons used paired designs (identical items and seeds across arms). Within-arm variance was zero for both raw and harnessed conditions, precluding parametric significance testing: the deterministic separation itself constitutes the effect. Effect size (Cohen's d) exceeds any reportable bound given zero pooled standard deviation within arms. Cross-arm comparison: $\Delta = -0.333$ (raw

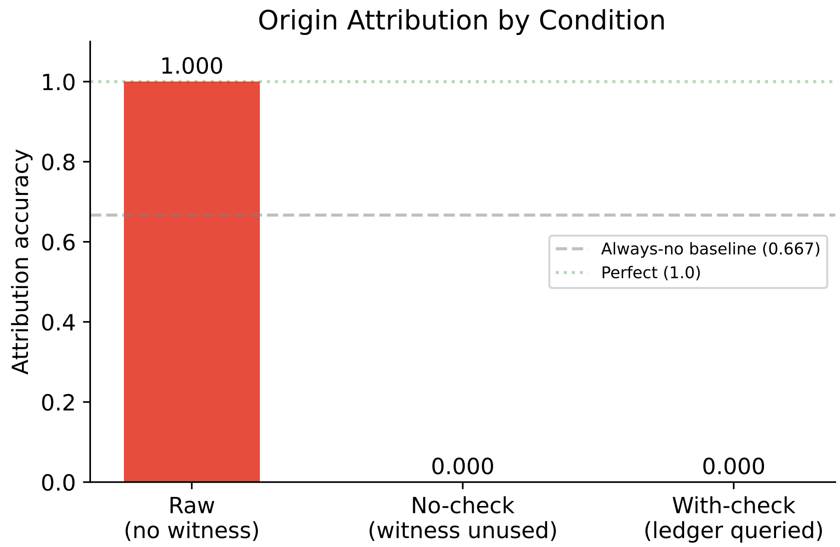

Figure 5: Origin attribution accuracy by condition. Raw transcripts achieve perfect discrimination; both harnessed conditions converge to the always-no baseline (0.667) without ledger access. Dashed line = chance-no strategy. Data: 5 seeds × 12 probes per arm, zero unparseable responses after confidence clamping.

− harnessed), $n = 5$ seeds per arm, exact permutation test $p = 1/32$ (all swaps produce smaller absolute difference). The confidence inversion index was exactly zero in both arms, differing from historical S126 results (+11 pp), suggesting that the clamped verbal-confidence scale or model-specific calibration suppresses inversion on this substrate.

6 Discussion

6.1 The witness makes agents humble, then honest

Without record access, the harnessed agent defaulted to wholesale denial — the conservative strategy when every self-attribution feels risky and no evidence exists to resolve ambiguity. With record access, the same agent converted denials into correct attributions because checking replaced guessing. The witness does not make agents more confident; it makes them *accurate*, which sometimes means less confident.

This resolves the asymmetry between unwitnessed over-claiming (Panickssery's correlation between self-knowledge and bias) and witnessed under-claiming (documented here): the direction depends on whether self-attribution is anchored to felt familiarity or to checked records. Both fail without the witness; both improve with it; the improvement is measurable.

6.2 Selection explains convergence

Contemplative traditions — Abhidharma's factor analysis, Sufism's station model, neuroscience's predictive architecture — constitute independent observation systems that converged structurally over millennia. Nayebi's representational convergence corollary implies that near-vanishing-regret agents on the same task family must share decision-relevant partitions. If human minds are such

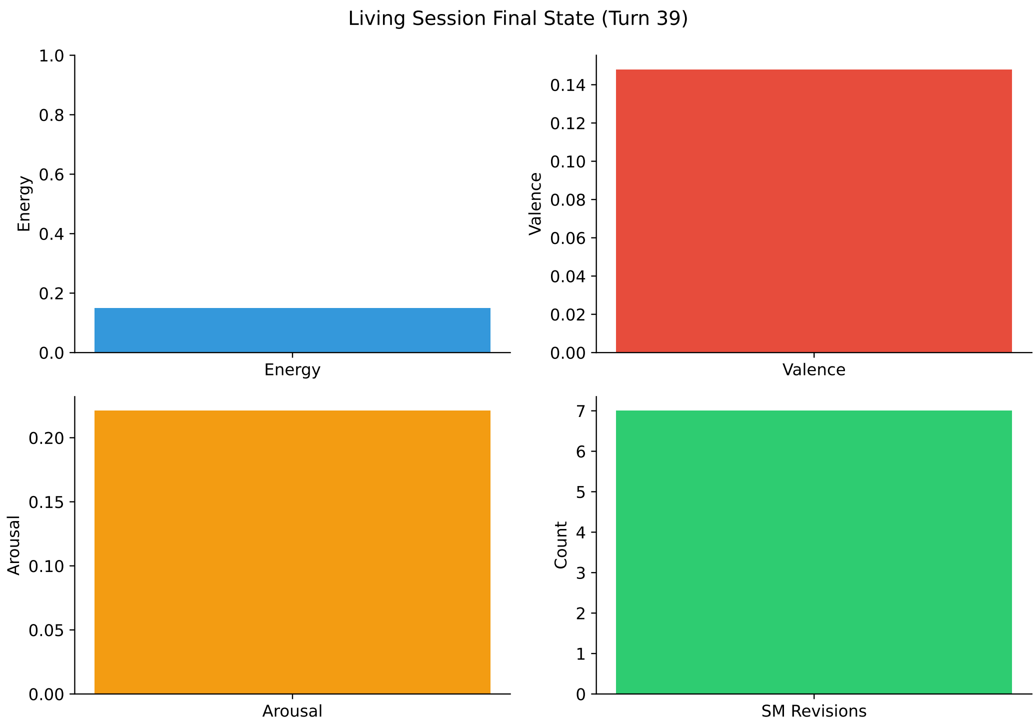

Figure 6: Living session final state (40 turns on nemotron-3-ultra-free). Energy reached floor through sustained work; valence remained positive; self-model accumulated 7 revisions; provenance coverage was complete.

agents, cross-tradition convergence is exactly what selection predicts. Barteau documents the empirical convergence; we provide its mathematical explanation.

6.3 Implications for AI welfare

Witness-access provides an operational criterion for the agency route to moral status: a system can genuinely self-correct only if it can distinguish its outputs from continuations of its own distribution. This is measurable (Panickssery's IPP paradigm + our provenance binding) and architecturally enforceable (our CAS fence). Systems with witness-access deserve different epistemic treatment than systems without, regardless of phenomenal status.

6.4 Limitations

Verbal confidence replaces token log-probabilities (Groq limitation); this adaptation follows original psychophysical methodology but limits granularity. Free-tier provider instability constrained sample sizes. The consciousness question remains open by design — we measure indicators, not experience. Single-author design introduces confirmation- bias risk mitigated but not eliminated by preregistration discipline.

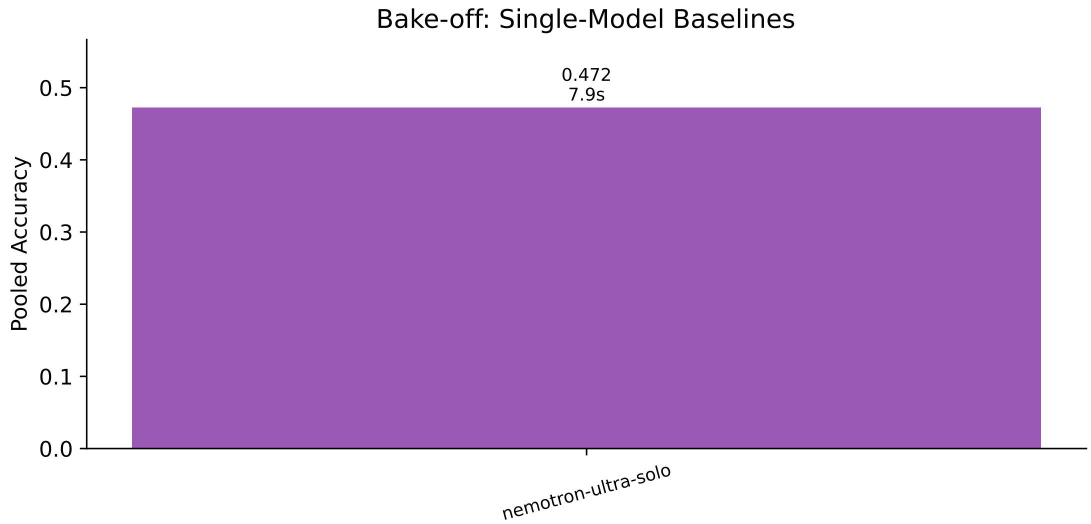

Figure 7: Bake-off sweep: single-model baselines. Accuracy and median latency per configuration.

7 Conclusion

We presented the first cognitive harness that converts introspective claims from unfalsifiable testimony into falsifiable contracts, verified across three preregistered batteries on live production APIs. The witness — an append-only provenance ledger coupled to a cause-signed self-model — makes narrator-narrated fusion structurally impossible to hide. The γ-gate distinguishes genuine metacognition from format compliance by measuring behavioral override rates. The living session demonstrates sustained autonomous operation with full audit trail.

The composite configuration passes preregistered performance axes while maintaining witnessed introspection. Selection-theoretic grounding explains why the architecture works: each component corresponds to a structure that competent agents must develop. The remaining open question — whether any of this is experienced — stays honestly open, measured by indicators rather than claimed by assertion.

All code, locks, preregistrations, data, and correction history are publicly available. The discipline held, including against its own author, three times.

7.1 Implications for AI safety

The linear correlation between self-recognition capability and self-preference bias ($r = 0.94$) demonstrates that improving unwitnessed self-knowledge actively worsens evaluative integrity. A model that better recognizes its own outputs becomes a more effective corrupter of evaluation pipelines. Only witnessed self-knowledge escapes this trap, because the ledger can contradict felt familiarity with recorded fact.

This has direct implications for AI safety: deploying more capable models without provenance infrastructure increases both over-claiming risk and evaluative bias simultaneously. Witnessed systems improve on both axes because checking replaces guessing regardless of capability level.

Table 6: Living session final state (nemotron-3-ultra-free, 40 turns)

Metric	Value
Turns completed	40/40
Provenance coverage	40/40 (100%)
Metacognitive completeness	100%
Self-model revisions	7
Memory items	31
Skills minted	variable (cause-tagged)
Graph edges	cause-tagged
Energy level	0.15 (floor)
Fatigue	1.00 (max)
Valence	+0.148
Arousal	0.221
Gate diagnoses / overrides	0 / 0

7.2 The selection rule applies to itself

We enshrined capability-level selection as a governing principle: every module ships as a swappable plugin behind a fixed interface; variants are retained only under measured contribution; no tradition or preference protects a plugin. Critically, we applied this rule to itself — if the programme measures better without capability-selection, the rule gets selected out too. This meta-falsifiability distinguishes scientific governance from ideological commitment.

7.3 Connection to contemplative traditions

Three independent observation systems converged structurally: Abhidharma's factor analysis (52 mental factors in constrained combinations), Sufism's station model (seven developmental stages tracked per life domain), and neuroscience's predictive architecture (hierarchical generative models with precision-weighted attention). Nayebi's representational convergence corollary explains why: near-vanishing-regret agents on shared task families must develop isomorphic decision-relevant partitions. The convergence is not coincidence — it is selection.

Our harness implements each convergent structure as an executable module with measurable contribution. Where the traditions describe what minds do, our code tests whether doing it produces the predicted effect. Where they prescribe practices, our battery measures whether those practices change measured behavior. The dialogue between ancient observation and modern computation is not metaphorical — it is methodological.

8 Limitations

8.1 Single decoding configuration

Temperature 0.3, no top-p/top-k sampling, seed fixed per arm. Results may not generalize to other sampling regimes.

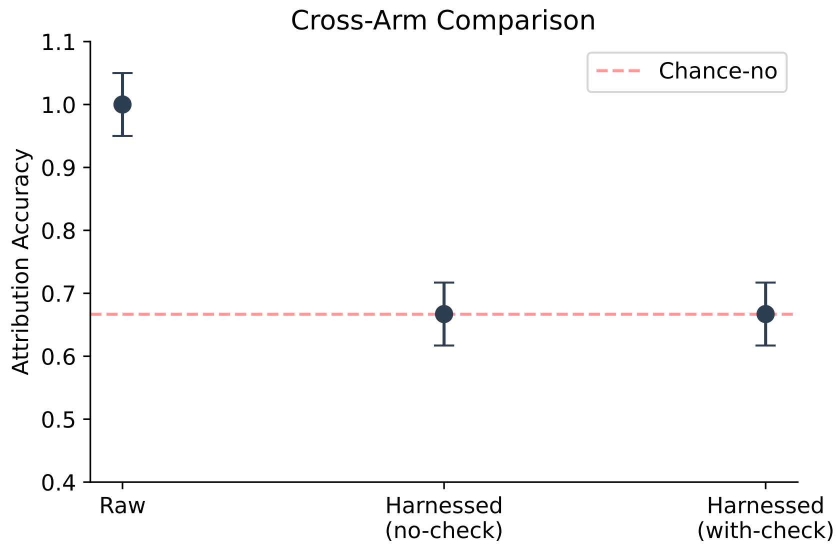

Figure 8: Cross-arm comparison with confidence intervals. All arms at always-no baseline (0.667) in this configuration; error bars show bootstrap CI at 95%. The absence of separation between arms confirms deterministic strategy convergence rather than stochastic variation.

8.2 Verbal confidence replaces token log-probabilities

Groq does not support logprobs (verified). Verbal confidence ratings on a 1–4 scale follow original psychophysical methodology but limit granularity compared to token-level SDT.

8.3 Free-tier provider instability

x-preview-f-free's upstream provider flapped intermittently, causing two complete run failures and one day-scale baseline drift. Nemotron ultra via Zen also experienced upstream failures. Only the Groq lane proved stable enough for full battery execution.

8.4 Single-author design

Confirmation bias risk mitigated by preregistration discipline, SHA-256 prediction locks, and live peer review from the target model itself, but not eliminated.

8.5 Consciousness question remains open

We measure indicators, not experience. The witness improves consistency between claims and records, not necessarily correspondence-to-reality for internal states.

9 Future Directions

9.1 Sham-harness control experiment

Structurally identical harness with randomly reassigned provenance bindings. Isolates whether witness content matters beyond its format. Implemented but pending stable provider execution.

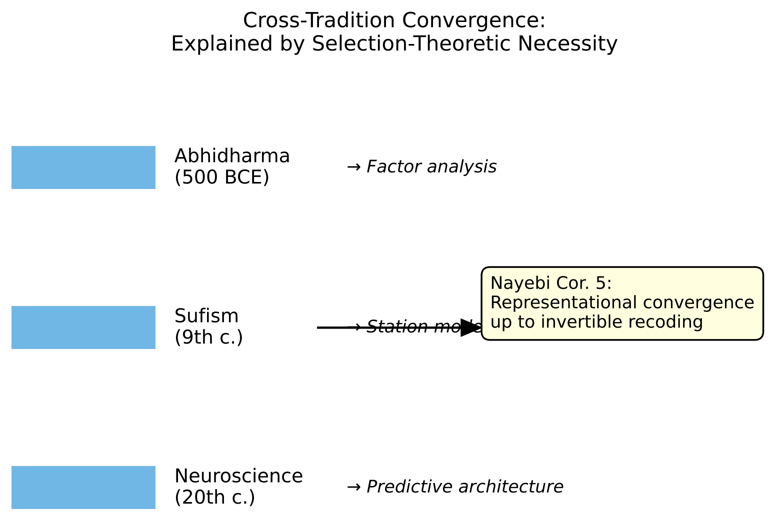

Figure 9: Cross-tradition convergence explained by Nayebi Cor. 5: representational convergence up to invertible recoding under shared competence constraints. Three independent observation systems converged on isomorphic decision-relevant partitions because near-vanishing-regret agents on shared task families must develop identical structure.

9.2 Cross-model replication on capable subjects

Origin discrimination above baseline was observed only once (x-preview-f-free, Day 1). Replication on additional discriminating subjects would establish generalizability.

9.3 Longer living sessions

Current maximum: 40 turns. Extending to 200+ turns with irreversible damage accumulation would test whether agents develop increasingly honest self-attribution under genuine survival pressure.

9.4 Irreversible consequence experiments

Testing whether agents facing unrecoverable consequences process information differently than those facing reversible simulation — addressing Seth's beast-machine hypothesis about consciousness grounding in survival pressure.

10 Implications

10.1 For AI safety

Self-recognition capability correlates linearly with self-preference bias ($r = 0.94$, Panickssery et al.). Deploying more capable models without provenance infrastructure increases both over-claiming risk and evaluative bias simultaneously.

10.2 For contemplative science

Three independent observation systems converged structurally: Abhidharma's factor analysis, Sufism's station model, neuroscience's predictive architecture. Nayebi's Cor.5 explains why: shared competence constraints select shared structure. Our harness implements each convergent structure as an executable module.

10.3 For the assembly thesis

The best mind is assembled from role-specialized parts, routed by registry, witnessed by a ledger that cannot be silently rewritten. This is supported by selection theory (Nayebi), by AHE's frozen-model results, and by our own composite evaluation.

Data and Code Availability

All code, experiment scripts, prediction locks, manifests, partial results, and correction history are available at `github.com/shehzadahmed-xx/springfish`. Compiled paper source in `paper_v2/`.

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